Learning Objective 9.1: I understand the process of decoupling and finding solutions for Maxwell's equations for dynamic fields.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Vector phasors.** In a lossless, source-free region, the electric field is

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\, 5\cos\left(\omega t - \beta z + 30^\circ\right) + \hat{\mathbf{y}}\, 3\sin\left(\omega t - \beta z\right) \text{ V/m}$$

   (a) Write the vector phasor $\widetilde{\mathbf{E}}(z)$.

   (b) Starting from $\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D}/\partial t$ in the time domain, derive the phasor form of the Ampere–Maxwell law in this region. State each assumption where you use it.

2. **Decoupling the magnetic field.**

   (a) Starting from the Ampere–Maxwell law, in phasor form, in a lossless, source-free region, derive the Helmholtz equation for the magnetic field. Name each of Maxwell's equations where you use it.

   (b) On a transmission line, which line quantities play the parts of μ and ϵ, and which plays the part of $\widetilde{\mathbf{H}}$?

3. **The Helmholtz equation.** A lossless dielectric has $\epsilon_r = 4$ and $\mu_r = 1$. The frequency is 300 MHz.

   (a) Find β, the wavelength, and the phase velocity.

   β =

   λ =

   v_p =

   (b) Show that $\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\, E_0 e^{-j\beta z}$ satisfies both the Helmholtz equation and $\nabla \cdot \widetilde{\mathbf{E}} = 0$.

   (c) The field $\widetilde{\mathbf{E}} = \hat{\mathbf{z}}\, E_0 e^{-j\beta z}$ also satisfies the Helmholtz equation. Explain why it cannot exist in a source-free region.

4. **The speed of the wave and the line.**

(a) Use $\mu_0 = 4\pi \times 10^{-7}$ H/m and $\epsilon_0 = 8.854 \times 10^{-12}$ F/m to find the speed of a wave in free space.

v_p =

(b) A coaxial line has $L' = 250$ nH/m and $C' = 100$ pF/m, and its dielectric has $\mu_r = 1$. Using $L'C' = \mu\epsilon$ for the coax (Lesson 30), find ϵ_r and the phase velocity.

ϵ_r =

v_p =

Documentation: _____